

## Lecture 12: Panel data II

**PPHA 34600**  
Prof. Fiona Burlig

Harris School of Public Policy  
University of Chicago

## From last time: regression-based DD

The simplest implementation of DD is just:

$$\hat{\tau}^{DD} = (\bar{Y}(\text{treat}, \text{post}) - \bar{Y}(\text{treat}, \text{pre})) - (\bar{Y}(\text{untreat}, \text{post}) - \bar{Y}(\text{untreat}, \text{pre}))$$

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To link these together, see:

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**Running this regression yields  $\hat{\tau} = \hat{\tau}^{DD}$**

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**In words:**

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- $E[\varepsilon_{it} | Treat_i, Post_t, X_{it}] = 0$

**In different words:**

- Conditional on covariates, treatment is as good as randomly assigned

**In other different words:**

- Treated and untreated units would be on similar trajectories if not for treatment



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**In different words:**

- Conditional on covariates, treatment is as good as randomly assigned

**In other different words:**

- Treated and untreated units would be on similar trajectories if not for treatment
- Just controlling for  $Treat_i$  and  $Post_t$  is usually not good enough!
- We also need a story for why  $Treat_i \times Post_t$  is quasi-random!

DD is a specific case of the fixed effects model, which lets us:

- Easily incorporate more units and time periods
- Allow for different units to be treated at different time periods
- Allow the effect of treatment to vary over time
- Enable us to more easily assess the identifying assumptions

# Fixed effects models

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- $\alpha_i$ : Individual-specific time-invariant bit
- $\delta_t$ : Common time-period-specific bit
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**Can we do better than leaving all of these in the error term?**

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## There are two main ways to do this:

- ① Use dummy variables
  - ② “De-meaning” the data
- These are both “fixed effects” estimators



## Fixed effects estimator 1: Use dummy variables

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$$Y_{it} = X_{it}\beta + \tau D_{it} + \sum_{i=1}^N \mathbf{1}[\text{unit} = i]_i + \nu_{it}$$

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- In general, define  $\tilde{V}_{it} = V_{it} - \frac{1}{T} \sum_{t=1}^T V_{it} = V_{it} - \bar{V}_i$

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We can then express the “de-meaned” version of our model:

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We can then express the “de-meaned” version of our model:

$$\tilde{Y}_{it} = \tilde{X}_{it}\beta + \tau \tilde{D}_{it} + \tilde{\nu}_{it}$$

→ This gives you the same result as “controlling for”  $\alpha_i$



## We can extend these estimators to a more complex model

Consider the regression model with both individual and time effects:

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De-meaning is more annoying. Now we de-mean by both time and unit:

Define

$$\begin{aligned}\tilde{V}_{it} &= V_{it} - \frac{1}{T} \sum_{t=1}^T V_{it} - \frac{1}{N} \sum_{i=1}^N V_{it} + \frac{1}{N \times T} \sum_{i=1}^N \sum_{t=1}^T V_{it} \\ &= V_{it} - \bar{V}_i - \bar{V}_t + \bar{\bar{V}}_{it}\end{aligned}$$

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The model is now:

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## Connecting FE to DD

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Now let's generalize:

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- $\alpha_i$  is a (set of) individual fixed effects, which captures  $\text{Treat}_i$
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- $\delta_t$  is a (set of) time fixed effects, which captures  $\text{Post}_t$
- Note that we've lost  $\alpha$ : this is now collinear with  $\alpha_i$
- With just two units,  $\alpha_i$  is just  $\mathbf{1}[\text{unit} = i]$  and  $\mathbf{1}[\text{unit} = j]$
- With just two time periods,  $\alpha_i$  is just  $\mathbf{1}[\text{time} = 0]$  and  $\mathbf{1}[\text{time} = 1]$

# Extending DD to multiple treatment times

What happens if we have treated units who get treated at different times?

- Our new general formulation works!

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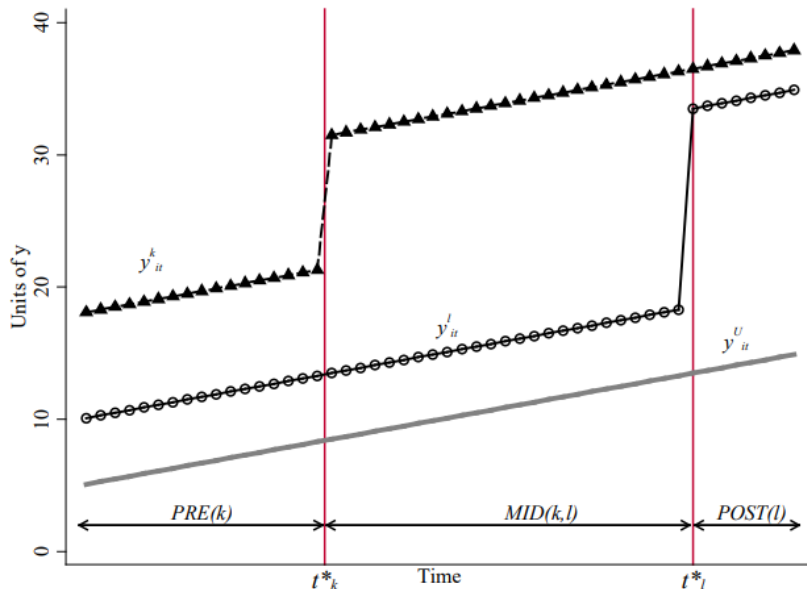
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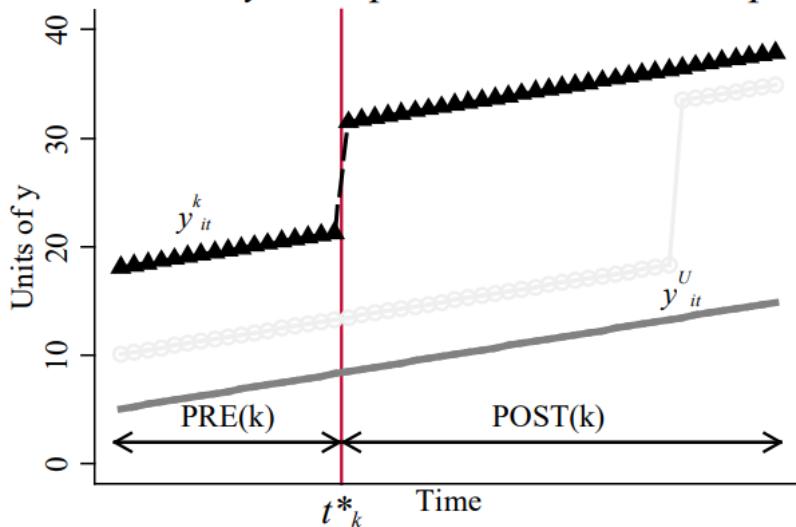
**Important note:** Running this specification gets you a **weighted average** of several comparisons. This may not be exactly what you want!

# What does multiple-treatment-timing FE get us?



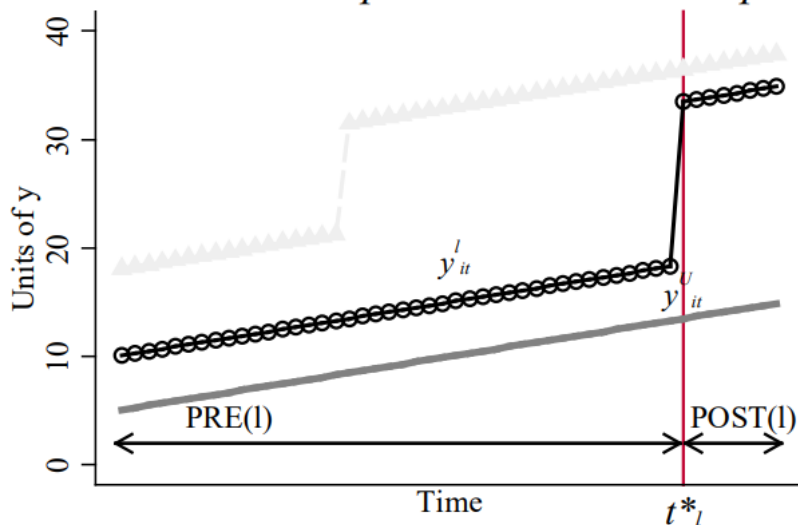
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## *A. Early Group vs. Untreated Group*



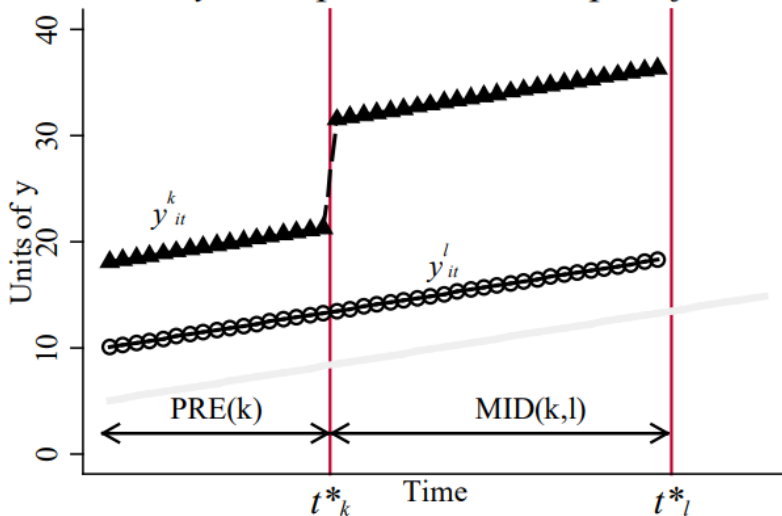
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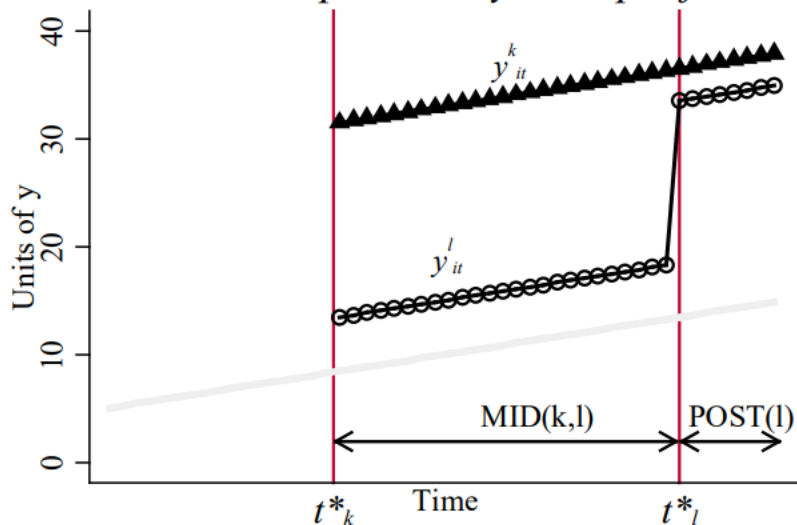
## C. Early Group vs. Late Group, before $t^*$



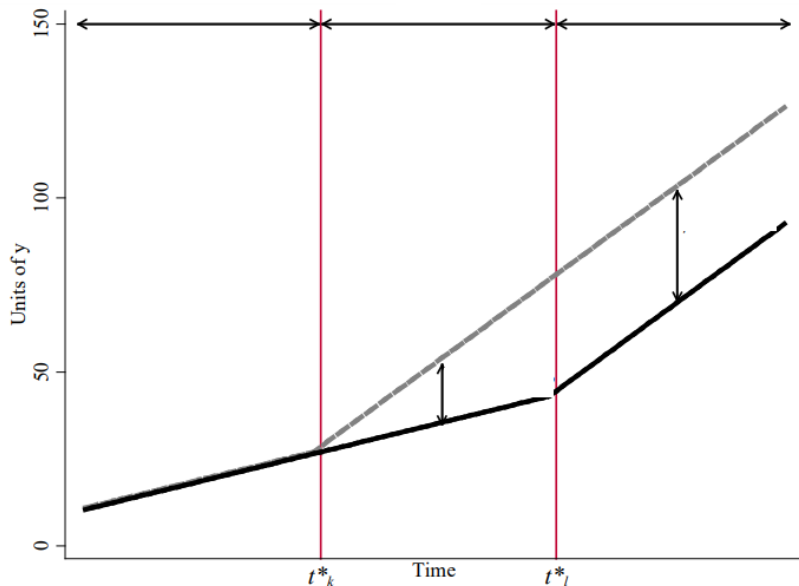


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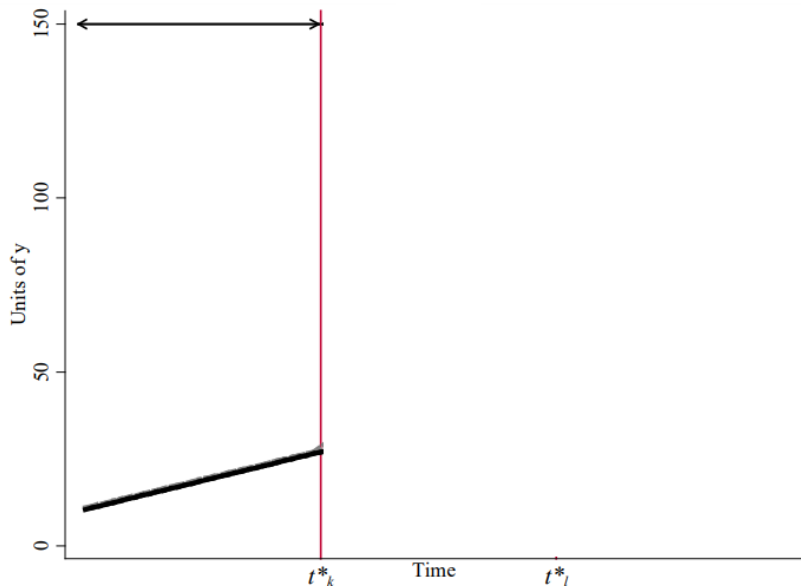
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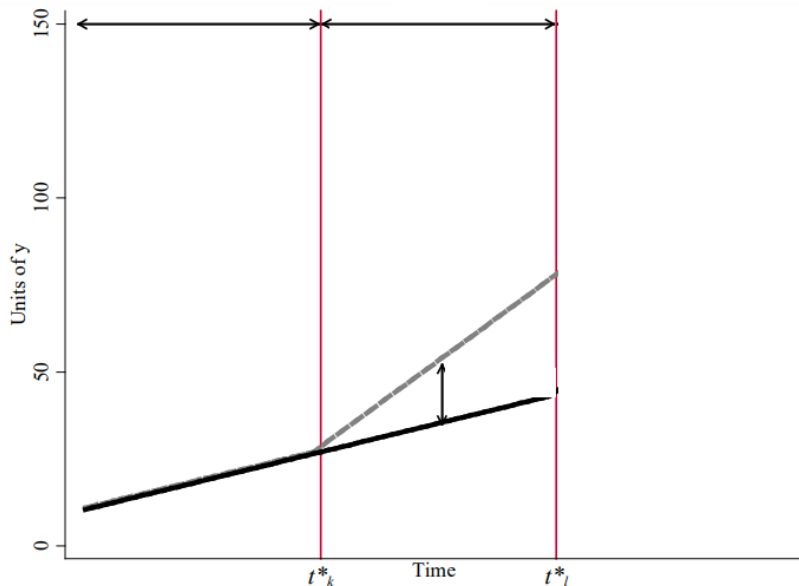
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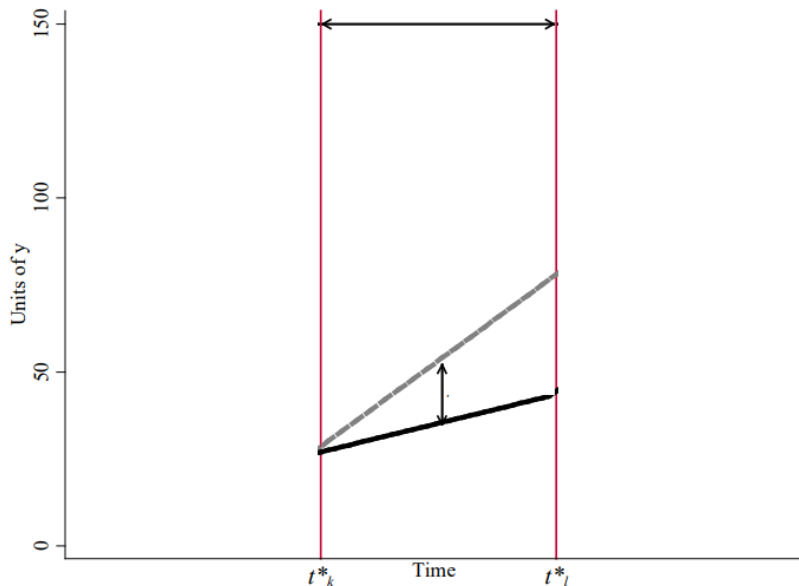
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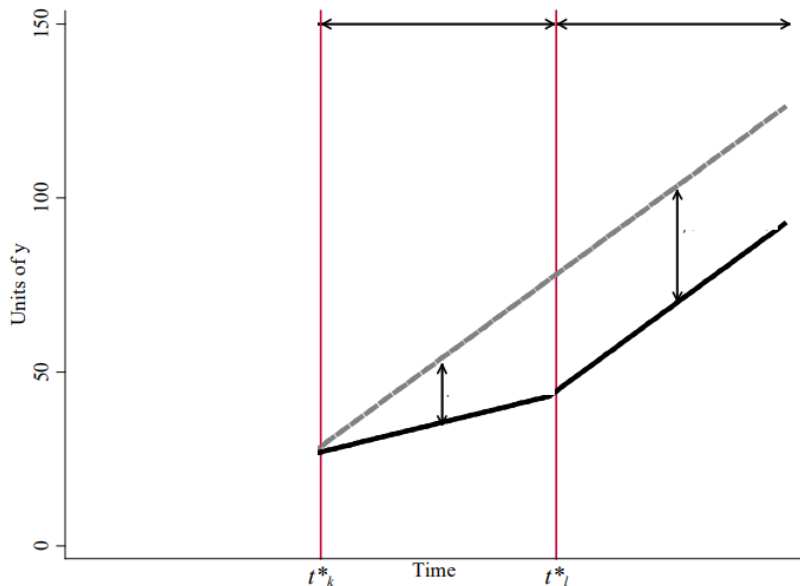
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# How do we do this right?

There are two main solutions to this problem:

- ① Weighted balance test to make sure this isn't a problem
  - Beyond the scope of this class
- ② Artificially “treat all units at the same time”

# The event study design

An event study is a more general FE design:

Our standard FE regression model:

$$Y_{it} = \tau D_{it} + \alpha_i + \delta_t + \beta X_{it} + \varepsilon_{it}$$

- This imposes the constraint that  $\tau_t = \tau$  for all  $t$
- (And all  $i$ , but that's a problem for a different day)



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A more general model will allow differential effects over time:

$$Y_{it} = \sum_{r=0}^R \tau_r D_i \times \mathbf{1}[\text{periods post treatment} = r]_{it} + \alpha_i + \delta_t + \beta X_{it} + \varepsilon_{it}$$

$\mathbf{1}[\text{periods post treatment} = r]_{it} = 1$  when we are  $r$  periods after treatment, 0 otherwise

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- The  $\tau_r$ s pick up the average treatment effect  $r$  periods after treatment

# The event study design

In the general version of this model, we include pre-treatment “effects”:

$$Y_{it} = \sum_{r=-S}^R \tau_r D_i \times \mathbf{1}[\text{periods to treatment} = r]_{it} + \alpha_i + \delta_t + \beta X_{it} + \varepsilon_{it}$$

- (Note the number of pre-treatment periods,  $S$ , and the number of post-treatment periods,  $R$ , need not be the same)

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$$Y_{it} = \sum_{r=-S}^R \tau_r D_i \times \mathbf{1}[\text{periods to treatment} = r]_{it} + \alpha_i + \delta_t + \beta X_{it} + \varepsilon_{it}$$

- (Note the number of pre-treatment periods,  $S$ , and the number of post-treatment periods,  $R$ , need not be the same)

What's so great about this design?

- This “lines up” treatment at the same time for everyone
- We can still use fixed effects to soak up confounders
- We get a (partial) test of the identifying assumption

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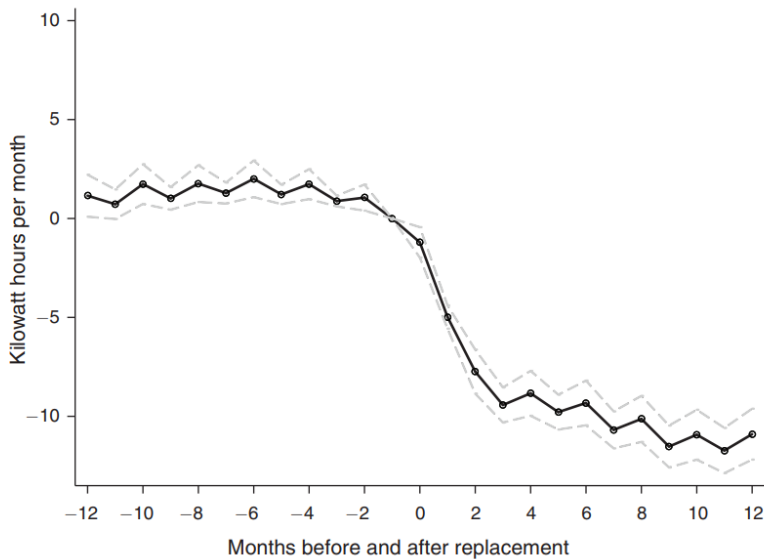
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- We can still use fixed effects to soak up confounders
- We get a (partial) test of the identifying assumption
  - We want pre-treatment  $\tau_s$ 's to be centered on 0 and not trending

## Remember this beautiful figure?



## TL;DR:

- ① We like the difference-in-differences approach a lot
- ② We discussed estimation with fixed effects
- ③ And covered the event study version

# For next class

## Topics:

- Panel data III

## Reading: Keiser and Shapiro (2018). You can skim:

- IV.C: Demand for water quality
- VII: Demand for water quality